SET 2

1. Design and configure Spanning Tree Protocol (STP) on a network with 2 switches connected redundantly using Packet Tracer, to prevent broadcast storms and loops.

2. Assume that you are the network administrator required to connect 40 systems across two buildings 200 metres apart. Choose the appropriate media and topology, and configure the network using Packet Tracer.

3. Using Wireshark, Capture and analyze the Packets that use BOOTP for different header fields and payload of the concerned packet.

4. Illustrate the concept of Subnetting – Class B Addressing for network address 172.16.0.0 and the subnet mask (255.255.255.0) (/24).

Set - 2.

1) Spanning Tree protocol (STP) using packet Tracer

Aim :-

To design and configure a network with 2 switches connected redundantly using Spanning Tree protocol (STP) to prevent broad cast storms and network loops.

Procedure :-

1) open Cisco packet Tracer
2) Add 2 switches (eg. 2960)
3) Add pcs and connect them to both Services.
4) connect the two switches using two links to create redudancy.
5) configure ip address for the pcs.
6) STP is enabled by default on Cisco switches.
7) Check STP status using the CLI
8) identify the Root bridge and Blocked port
9) Test communication between pcs.

Configuration Steps

Switch 1:

```
enable
configure terminal
hostname sw1
spanning tree mode pvst
spanning tree vlan 1 root primary
end
show spanning - tree
```

Switch 2 :

```
enable
configure terminal
hostname Sw2
spanning-tree mode pvst
spanning-tree vlan 1 root secondary.
end
show spanning tree.
```

Result :-

STP was successfully configured the redundant link was placed in blocking state, preventing network loops and broadcast storm while maintaing network connectivity.

2) Network for 40 System in Two Buildings :-

Aim :-

To design and configure a network connecting 40 system across two building located 200 metres apart, using suitable transmission media and topology in Cisco packet tracer.

Procedure :-

1. Open Cisco packet tracer
2. Create Building 1 and Building 2
3. Add 20 pcs in each building
4. Add switches to connect the pcs
5. Since the building are 200 m apart, use fiber optic cable for inter building connection.

6. connect pcs to switches using copper Ethernet using fiber.

7. connect pcs to switches btw the buildings using fiber.

8. Assign IP address to all pcs

9. Test connectivity using the ping command

Configuration steps :-

Building 1 :

192.168.1.1 – 192.168.1.20

192.168.1.21 – 192.168.1.40.

Building 2 :

192.168.1.21 – 192.168.1.40

Subnet mask :-

255.255.255.0.

Example PC configuration :-

IP Address : 192.168.1.1

subnet mask : 255.255.255.0

Media :

→ pc → switch : copper straight - through

→ switch → switch between building : fiber optic.

Result :-

A network connecting 40 system in two buildings was successfully designed and configured - fiber - optic cable was used for the 200 m inter building connection, and communication between system was successfully verified.

3. BOOTP packet Analysis packets using wire shark.

Aim :-

To capture and analyze BOOTP packets using wire shark and study their header fields and pay load.

Procedure :-

1. install and open wireshark.
2. Select the active network interface.
3. start packet capture
4. Generate network traffic that use DHCP/ BOOTP.
5. stop the capture
6. apply the filter. bootp.
7. Select a BOOTP/DHC section.
8. analyze the header fields and payload.

Configuration :-

No special configuration is required in Wireshark. Use the display filter :- bootp.

Result :-

BOOTP packets were successfully captured and their header fields and payload were analyzed using Wireshark.

4. Class B Subnetting :

Aim :

To demonstrate class B subnetting for the network address 172.16.00. using the subnet mask 255.255.255.0 (/24)

Procedure :-

1. Identify the given class B network address 172.16.0.0 using the subnet mask 255.255.255.0 (/24).

2. Use the subnet mask 255.255.255.0

3. Calculate the number of subnets and hosts.

4. Identify the network, usable host and broadcast address

5. Assign the required subnet address

Configuration :-

Network : 172.16.0.0/24

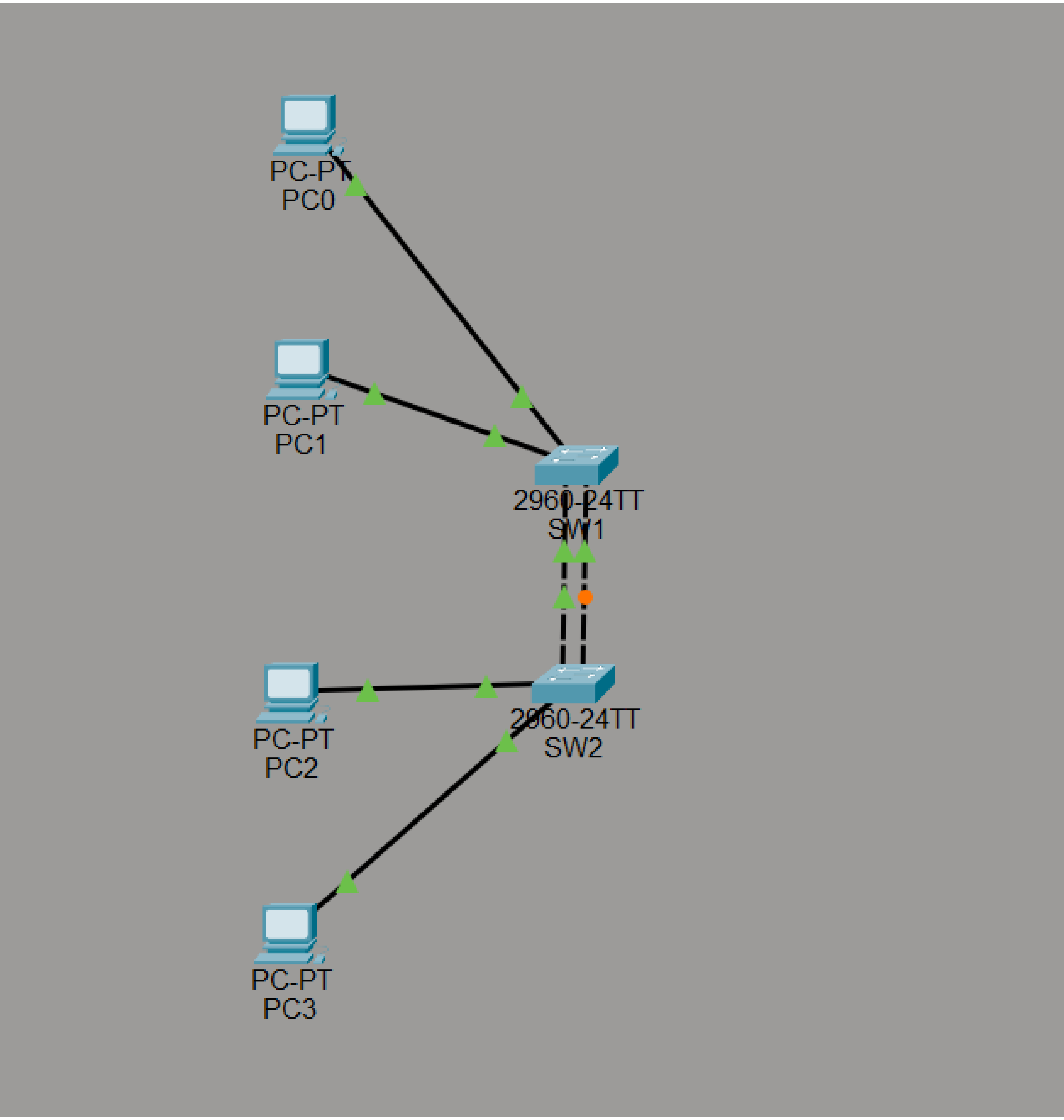
- Subnet mask. 255.255.255.0.
- number of /24 subnets from the original /16 : 256
- usable hosts per subnet : 254.

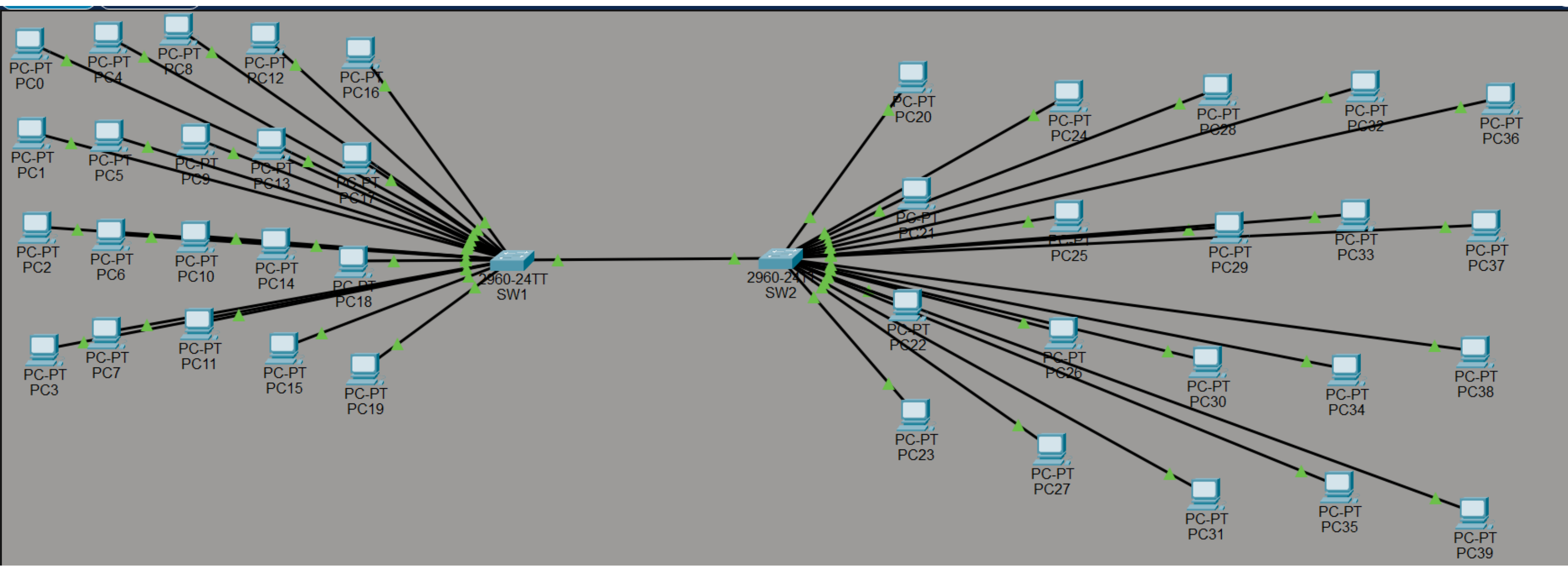
Example :-

Network	usable Hosts	Broad.
172.16.0.0	172.16.0.1 - 172.16.0.254	172.16
172.16.1.0	172.16.1.1 - 172.16.1.254	172.16
172.16.2.0	172.16.2.1 - 172.16.2.254	172.16

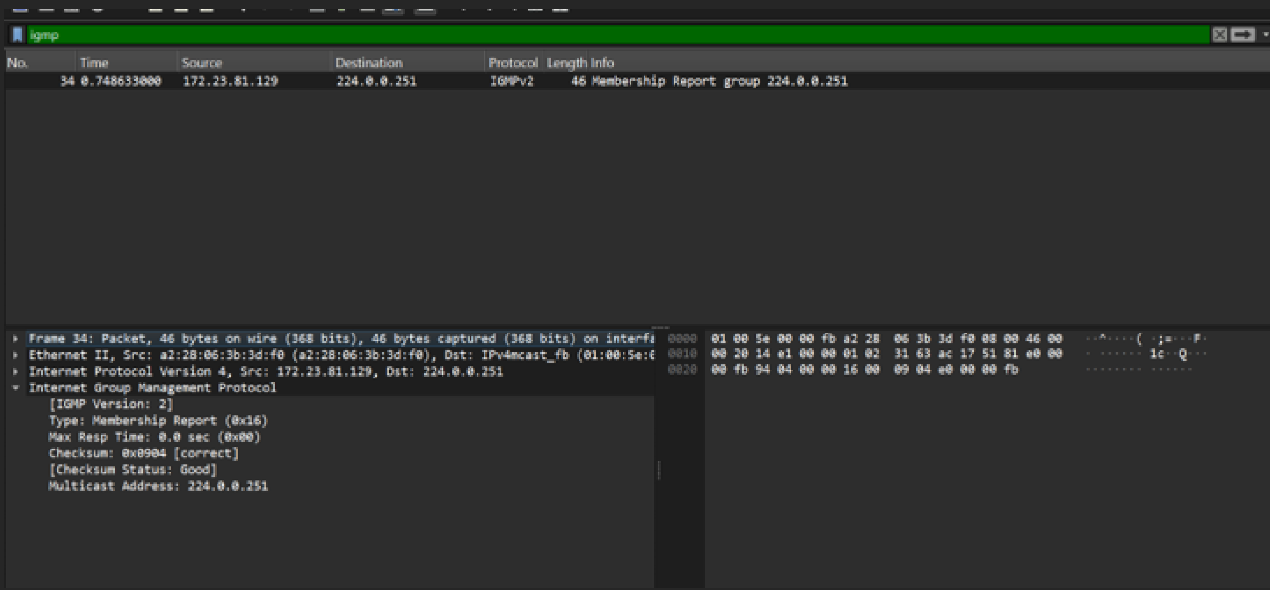
Result :-

The Class B network 172.16.0.0 was sucessfully subnetted using the /24 subnet mask, provide 256 subnets with 254 usable hosts per subnet.


PC-PT
PC0
PC-PT
PC1
2960-24TT
SW1
PC-PT
PC2
2960-24TT
SW2
PC-PT
PC3



PC-PT PC0
PC-PT PC4
PC-PT PC8
PC-PT PC12
PC-PT PC16
PC-PT PC1
PC-PT PC5
PC-PT PC9
PC-PT PC13
PC-PT PC17
PC-PT PC2
PC-PT PC6
PC-PT PC10
PC-PT PC14
PC-PT PC18
PC-PT PC3
PC-PT PC7
PC-PT PC11
PC-PT PC15
PC-PT PC19
2960-24TT SW1
2960-24TT SW2
PC-PT PC20
PC-PT PC24
PC-PT PC28
PC-PT PC32
PC-PT PC36
PC-PT PC21
PC-PT PC25
PC-PT PC29
PC-PT PC33
PC-PT PC37
PC-PT PC22
PC-PT PC26
PC-PT PC30
PC-PT PC34
PC-PT PC38
PC-PT PC23
PC-PT PC27
PC-PT PC31
PC-PT PC35
PC-PT PC39



Field	Value
Protocol	**IGMPv2**
Type	**Membership Report (0x16)**
Source IP	**172.23.81.129**
Destination IP	**224.0.0.251**
Max Response Time	**0.0 sec**
Checksum	**0x0904 — Correct**
Multicast Address	**224.0.0.251**
Packet Length	**46 bytes**

implemented the Stop-and-Wait protocol using C. The sender sends one frame and waits for an acknowledgement. If the acknowledgement is not received, the frame is retransmitted. After receiving the ACK, the sender proceeds to the next frame

```
Moham@Desktop UCRT64 ~
# /c/Users/Moham/stopwait.exe
Enter number of frames: 3

Sending Frame 1...
Enter ACK for Frame 1 (1 = received, 0 = lost): 1
ACK received for Frame 1
Frame 1 transmitted successfully.

Sending Frame 2...
Enter ACK for Frame 2 (1 = received, 0 = lost): 0
ACK not received for Frame 2
Retransmitting Frame 2...
Frame 2 transmitted successfully after retransmission.

Sending Frame 3...
Enter ACK for Frame 3 (1 = received, 0 = lost): 1
ACK received for Frame 3
Frame 3 transmitted successfully.

All frames transmitted successfully.

Moham@Desktop UCRT64 ~
# |
```